Lucky Goyal

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Professional Summary

AI Engineer with hands-on experience in Generative AI, NLP, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), vector databases, and transformer-based model fine-tuning. Experienced in building end-to-end AI pipelines using Python, Hugging Face, FLAN-T5, Qdrant, and vector search for practical NLP applications. Strong foundation in Machine Learning, Deep Learning, Data Structures & Algorithms, and software engineering, with a 4th-place achievement in the BEA 2019 GEC Competition.

Technical Skills

Languages & Databases: Python, Java, JavaScript (ES6+), SQL, HTML5, CSS3, MongoDB, Qdrant (Vector DB)
AI / ML & Data Science: Machine Learning, Deep Learning, Neural Networks, NLP, Generative AI, LLMs, RAG Architecture, Hugging Face, FLAN-T5, TensorFlow, Keras, OpenCV, MediaPipe, Prompt Engineering, Agentic AI
Full Stack & Frameworks: React.js, Node.js, Express.js, RESTful APIs, JWT Authentication
Developer Tools & Practices: Git, GitHub, VS Code, Data Preprocessing, Model Evaluation

Professional Experience

Next Education India Private Limited June 2026 – Present
AI Engineer Intern *Hyderabad, India*

- Researched, benchmarked, and evaluated commercial and open-source Grammatical Error Correction (GEC) models and datasets, conducting licensing analysis, data preprocessing, and comparative feasibility studies for model fine-tuning.
- Selected and fine-tuned FLAN-T5-Large for GEC, optimizing target-generation accuracy and adapting the model to domain-specific grammatical correction tasks.
- Architected and implemented a practical Retrieval-Augmented Generation (RAG) pipeline using Qdrant Vector Database, automated API authentication, vector collections, and synthetic CSV data ingestion in Python.
- Participated in the BEA 2019 GEC Shared Task, submitting the developed GEC model and achieving 4th place among participating systems.
- **Technologies:** Python, Hugging Face, FLAN-T5, RAG, Qdrant, NLP, Data Preprocessing, Git.

Key Projects

AI Chatbot Assistance for Visually Impaired | *Python, OpenCV, YOLO, GenAI, NLP*

- Engineered an accessible multimodal AI assistant integrating Speech-to-Text (STT), Text-to-Speech (TTS), and real-time computer vision for scene understanding.
- Deployed cloud-integrated Generative AI and YOLO object detection models to deliver real-time audio guidance and environmental perception, enhancing user independence.

Real-Time Indian Sign Language (ISL) Recognition | *TensorFlow, Keras, CNN, OpenCV, MediaPipe*

- Constructed a deep learning pipeline leveraging Convolutional Neural Networks (CNN) and MediaPipe hand tracking to recognize and translate sign gestures into text in real time.
- Processed video frames using OpenCV and achieved high classification accuracy across dynamic gesture datasets.

Worknest System – Enterprise Workplace Management | *React.js, Node.js, Express.js, MongoDB, JWT*

- Developed a full-stack MERN enterprise portal managing employee records, attendance tracking, leave workflows, and dynamic task assignments.
- Engineered Role-Based Access Control (RBAC), secure JWT authentication, and high-throughput RESTful CRUD APIs with MongoDB database optimization.

Education

GLA University Aug 2022 – June 2026
*Bachelor of Technology (B.Tech) in Computer Science – **CGPA: 7.8/10.0*** *Uttar Pradesh, India*

Kanha Makhan Public School Apr 2020 – May 2021
*CBSE Class 12 (AISSCE) – **Score: 83.6%*** *Uttar Pradesh, India*

Certifications & Achievements

- **4th Position – BEA 2019 GEC Competition:** Participated in the BEA 2019 Grammatical Error Correction (GEC) Shared Task, submitting a fine-tuned GEC model and securing 4th position among participating systems.
- **1st Position – IBM ICE Day Event:** Secured 1st place in a Technical Paper Presentation focused on innovative software solutions and emerging technologies.
- **Top Performer – NPTEL Certification:** Achieved 1st Position / Top Performer in Organizational Behaviour: Individual Dynamics in Organization, conducted by IIT Guwahati.
- **Competitive Programming:** Active LeetCode problem solver with consistent practice in Data Structures & Algorithms, problem-solving, and competitive programming.